

RSA Digital Signature Scheme.

Design

1. Alice Select p, q
as large prime no.s
2. Calculate $n = p \times q$
3. Calculate $\phi(n_A) = (p-1) \times (q-1)$
4. Calculate public key
 e_A Such that $\gcd(e_A, \phi(n_A)) = 1$
5. Calculate private key
 $d_A = \frac{1 + (k \times \phi(n_A))}{e_A}$

Bob Select p, q
as large prime no.s

Calculate $n = p \times q$

Calculate $\phi(n_B) = (p-1) \times (q-1)$

Calculate public key.

e_B ; Such that $\gcd(e_B, \phi(n_B)) = 1$

Calculate private key

$d_B = 1 + \frac{(k \times \phi(n_B))}{e_B}$

Signature

Consider a message w such that $w < n$

Signature of w is (w, σ) where $\sigma = w^d \bmod n$.

Verification

Verification of Signature is $\text{Verify}(w, \sigma)$

where $w = \sigma^e \bmod n$.

Example

1. Alice $p=7$; $q=13$
2. Select $n = 7 \times 13 = 91$
3. $\phi(n) = (p-1) \times (q-1) = (7-1) \times (13-1)$
 $= 6 \times 12 = 72$
4. Select $e_A = 5$; $\gcd(5, 72) = 1$
5. Select $d_A = \frac{1 + (k \times \phi(n))}{e_A}$
 $= \frac{1 + (k \times 72)}{5}$ $k=2$; $d_A = 29$.

Public key $e_A = 5$; private key $d_A = 29$.

~~Signature~~ Signature

Select w such that $w < n$

$$w = 35$$

$$\begin{aligned}\text{Signature}(w, \sigma) &= \sigma = w^d \bmod n \\ &= 35^{29} \bmod 91 = 42.\end{aligned}$$

Verification

$$\text{verify}(w, \sigma) = w = \sigma^e \bmod n$$

$$= 42^5 \bmod 91 = 35.$$